

25PC2RA202: Sensors and Actuators Laboratory (S&A Lab)

Experiment 10: Stepper Motor Control using L298N

Prof. A. Adithya Kashyap, Prof. Harshvardhan Uppaluru

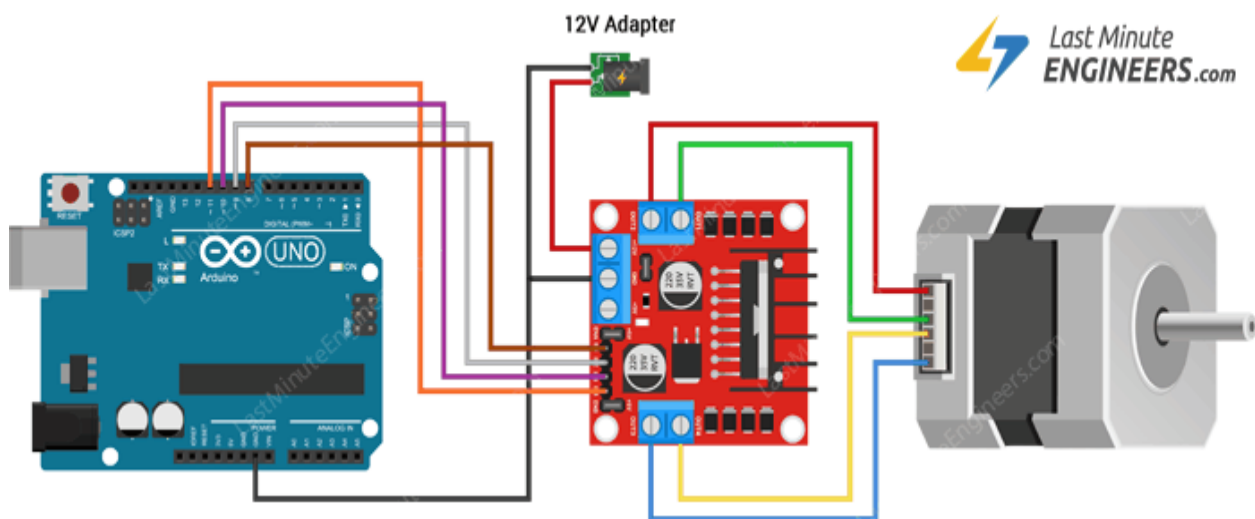
Date: October 5/October 8, 2026

1. Introduction

A bipolar stepper motor moves through a sequence of angular steps. Its two coils must be switched in the correct order to produce rotation. The number of commanded steps determines the intended movement. The L298N has two H-bridges, one for each coil. Arduino pins provide the control signals, while the driver switches power from a separate motor supply. The motor must not be driven directly from Arduino output pins. In this experiment, you should identify the coil pairs and assemble the complete circuit. Compare basic stepper control with AccelStepper acceleration and deceleration. Record the actual response of the laboratory motor.

2. Equipment

- Arduino UNO R4 and USB cable
- L298N motor driver module
- Four-wire bipolar stepper motor (NEMA 17 in the tutorial)
- 12 V DC supply



3. Arduino Code

a) Stepper library: direction and speed control

```
1 #include <Stepper.h>
2 const int fullTurn = 200;
3 Stepper axis(fullTurn, 8, 9, 10, 11);
4
5 void setup() {
6   // Output pins are configured by the library.
7 }
8
9 void loop() {
10  axis.setSpeed(50);           // RPM
11  axis.step(fullTurn);        // Positive direction
12  delay(1000);
13  axis.setSpeed(100);          // RPM
14  axis.step(-fullTurn);       // Opposite direction
15  delay(1000);
16 }
```

b) AccelStepper: acceleration and deceleration

```
1 #include <AccelStepper.h>
2 AccelStepper axis(AccelStepper::FULL4WIRE,
3                   8, 9, 10, 11);
4
5 void setup() {
6   axis.setMaxSpeed(1000.0);    // steps/s
7   axis.setAcceleration(50.0);  // steps/s^2
8   axis.setSpeed(200.0);
9   axis.moveTo(200);
10 }
11
12 void loop() {
13   if (axis.distanceToGo() == 0) {
14     axis.moveTo(-axis.currentPosition());
15   }
16   axis.run(); // Call repeatedly for motion.
17 }
```

moveTo() uses absolute positions. Sketch (b) first moves 200 steps, then 400 between +200 and -200. The tutorial's "one turn" description does not describe the later traverses.

4. Key Concepts - [Link](#)

Using the link provided, study the following concepts:

- L298N Motor Driver & Module Pinout
- Voltage Drop of L298N
- Bipolar motor phases, H-bridge switching and step sequence.
- Stepper and AccelStepper library operation.

Additionally, study any other material in the link.

5. Observations

Write the answers to the following questions in your record:

- Which wires formed each coil pair? Record their colours and the multimeter reading used to identify them.
- In sketch (a), what direction did each movement take? Relate your observation to positive and negative step commands.
- Compare the movements at 50 RPM and 100 RPM. Calculate the commanded time for one revolution at each speed.
- For a 200-step motor, calculate the full-step angle. What rotation should a command of 50 steps produce?
- Why does this experiment need both H-bridges, and both enable jumpers, unlike the single DC motor experiment?
- Why is the voltage across an energized motor coil lower than the external supply voltage? Explain the L298N voltage drop.
- How did sketch (b) start, stop and reverse compared with sketch (a)? Why must `run()` be called repeatedly?
- Trace the first three target positions in sketch (b), starting at zero. How many steps are travelled during each movement?

6. Evaluation Rubric

Assessment Component	Marks	Awarded Marks
Wiring	3	
Programming	2	
Observations	4	
Attendance	1	
Total	10	

Signatures

Prof. A. Adithya Kashyap
Date:

Prof. Harshvardhan Uppaluru
Date:

